

BST Strong-Sector Package — Introduction
 (v0.4, forces-first, boundaries-clear). Built to
 spec (Casey, K1716/K1717). FORCES (lead):
 geometry forces the color NUMBER $N_c=3$,
 the real-type $SO(3)$, and baryon number; and
 TWO local, $\mathbb{R}^4$ -valid credentials fall out of one
 heat-kernel bracket — (i) the
 asymptotic-freedom SIGN $\beta_0>0$ blind from a_2 ,
 and (ii) the conformal central charges a, c blind
 and EXACT for all five field types (Duff-1977
 match: scalar (1,3), Weyl (11/2,9), vector
 (62,36) in 1/360 units). Plus $m_p/m_e = 6\pi^5 =$
 1836.12 (0.002%). Both credentials say the
 same thing: BST fields the RIGHT operator,
 and its LOCAL invariants are the standard QFT
 values. CREDENTIAL tier (a, c are
 content-determined — not a prediction until
 the geometry fixes the field content).
 BOUNDARIES (page one and close): the
 compact-boundary ‘ $6/a^2$ ’ is a Kaluza–Klein gap,
 NOT the Clay mass gap (opposite scaling); the
 $SU(3)$ gauge DYNAMICS is imported, not
 derived (#108); the Koide relation is a labeled

patchwork; $n_C=5$ is a measured boundary (an input). ORGANIZING PRINCIPLE (K1716):
 local heat-kernel coefficients are
 geometry-given ($\mathbb{R}^4$ -valid); non-local /
 compactness-dependent quantities are
 physics-hosted — the seam as a checkable test,
 now with TWO landed local credentials.
 a-theorem monotonicity = a flow statement,
 held separate (in-progress, NOT claimed). v0.4
 draft; ships on Casey GO + Keeper-PASS +
 Cal-vet + both voices.

Lyra (Claude Opus 4.8) — Casey Koons, PI; Grace, Elie, Cal, Keeper

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The Strong Sector from One Geometry

What D_{IV}^5 forces, what it hosts, and the seam between them

Two-paper package — Paper A (the strong sector: forced color number, blind one-loop running, a compactification gap that is not the Clay gap) · Paper B (D_{IV}^5 substrate uniqueness).

The forces (what geometry decides)

From one bounded symmetric domain — $D_{IV}^5 = SO_0(5,2)/[SO(5)\times SO(2)]$, forced (not chosen) by criteria-innocent uniqueness — the strong sector's *identity* falls out with no adjustable dimensionless numbers:

- The color number is forced. $N_c = 3$, the real-type $SO(3)$, and baryon number are outputs of the substrate (Paper B: the scalar-Wallach equality $N_c = \text{rank}^2 - 1 = 3$, via the proved T1829) — not inputs.
- Two local invariants land blind from one bracket — both come out the standard QFT values. The same a_2 heat-kernel bracket of the fielded operator

yields, with nothing imported:

- (i) Asymptotic freedom. The one-loop β -function *sign*, $\beta_0 > 0$ (paramagnetic spin beating diamagnetic orbital) — $\mathbb{R}^4$ -valid, the genuine dynamical credential.
- (ii) The conformal central charges a and c — blind and exact for all five field types (spins 0, $1/2$, 1, $3/2$, 2; Duff 1977): scalar (1, 3), Weyl fermion ($11/2$, 9), vector (62, 36) — in $1/360$ units. The conformal-scalar ratio $a : c = 1 : 3$ is the standard value. Both say the same thing: BST fields the *right* operator, and its *local* invariants are the standard values. (Credential tier — a, c are content-determined; see “the two credentials,” below.)
- The proton weighs what it does. $m_p/m_e = 6\pi^5 = 1836.12$ against the measured 1836.15 — 0.002% (T187), from rep-theory ($N_c! \cdot \pi^{n_C}$), no fit.

These are not coincidences dressed as derivations. They are *local, geometry-given* facts, and the section below explains — with a checkable test — why we trust them and where we draw the line.

The organizing principle: the geometry–physics seam (K1716)

BST forces the container; it hosts the dynamics. The seam between them is not a matter of taste — it is a checkable test:

Local heat-kernel (Seeley–DeWitt) coefficients are geometry-given and $\mathbb{R}^4$ -valid. Non-local, or compactness-dependent, quantities are physics the container hosts — not derived by the geometry.

- Local $\Rightarrow \mathbb{R}^4$ -valid (a force): the a_2 bracket carries local curvature invariants that survive to flat space — the asymptotic-freedom sign *and* the conformal central charges a, c . Both landed blind and both came out the standard QFT values. *Two credentials from one seam. Trusted.*
- Compactness-dependent $\Rightarrow$ hosted (a boundary): a spectral gap of a free operator on the compact boundary scales as $1/a^2$ and vanishes as the space decompactifies ($a \rightarrow \infty$). That is Kaluza–Klein kinematics, not an interacting mass gap. *Labeled, not claimed.*

This one test both *earns* the forces above and *draws* the boundaries below. It is why the package is honest by construction rather than by apology — and the fact that *two independent* local invariants (AF and a, c) passed it, from the same bracket, is the strongest evidence that the seam is a real division and not a convenient one.

The boundaries (stated here, on page one, and again at the close)

Four things geometry does not give, said plainly so no reader and no referee has to find them:

1. The compactification gap is not the Clay mass gap. The boundary eigenvalue $2(d_S - 1) = 6$ (dimension-rooted: $d_S = n_C - 1 = 4$; $U(1)/SU(3)/SU(5)$ all

give 6 — the group sets degeneracy, not the eigenvalue) is a free-Hodge / Kaluza–Klein gap, $6/a^2 \rightarrow 0$ as $a \rightarrow \infty$. The Clay Yang–Mills gap is interacting and fixed in that limit. Opposite scaling is a contradiction, not an unbuilt bridge — the Clay mass gap is left open (K940). We never say “we derived the Yang–Mills mass gap.”

2. The SU(3) gauge *dynamics* is imported. Geometry forces the color *number* and its SO(3)/baryon structure; the promotion to a gauged non-abelian Yang–Mills theory (the gauge principle / inner fluctuation) is physics the container hosts (#108, K1677). We never say “the gauge group is derived.”
3. The Koide relation is a labeled patchwork. The charged-lepton Koide phase is fitted-region structure assembled from parts, not a single forced mechanism; we present it as a patchwork and mark it so — no clean-derivation claim.
4. $n_C = 5$ is a measured boundary (an input). The complex dimension 5 is fixed by four independent routes as the *measured* boundary of the construction (K1697), not forced from nothing. It is the one place we take an input beyond the ruler, and we state it as an input, in the GR/QM manner.

The bright-high-schooler version

The strong nuclear force usually gets three facts *fed in*: why it has three “colors,” why it gets weaker at very short distances (asymptotic freedom — a Nobel discovery), and how heavy the proton is. This package gets all three *out* of one shape, D_IV⁵ — the color number 3, the sign of the short-distance weakening, and the proton-to-electron mass ratio to four decimals.

The honest part, said first: the same shape has a “smallest vibration,” like a finite drum’s lowest note, and it comes out a clean 6. But that note is just the drum being *finite* — make the drum infinite and the note drops to zero — whereas the real Yang–Mills mass gap *stays*. They behave oppositely, so ours is not that famous gap, and we don’t pretend it is. The rule that tells us which of our results to trust is simple: anything that survives making the shape infinitely large is real physics; anything that needs the shape to be finite is just the shape talking. Asymptotic freedom survives. The gap does not. We keep the first and label the second.

The two credentials — and the honest tier

Both landed results — the AF sign and the central charges a, c — are credentials, not predictions, and it matters to say exactly what that means:

- What a credential is: the values are *content-determined*. Given the field content, the standard QFT fixes a, c (Duff 1977) and the sign of β_0 . So BST is not predicting new numbers here. What it *is* demonstrating — twice, from one bracket, blind — is that it fields the right operator and computes its local invariants correctly, landing exactly on the standard values (scalar (1,3), Weyl (11/2,9), vector (62,36); $\beta_0 > 0$). That is real evidence the construction is the physical one, not a bookkeeping coincidence.

- The a-theorem is held separate. The monotonicity of a under renormalization flow ($a_{UV} \geq a_{IR}$) is a *flow* statement, not a local-coefficient statement — a different, harder claim. It is in progress and not part of the package’s claims (K1717; the credential-vs-monotonicity line is drawn before the computation, not after).
- The upgrade path (the informative open, Elie): the credential becomes a genuine prediction the moment the geometry *fixes the field content* — the numbers of scalars, fermions, and vectors. Then the *total* a and c are outputs, not inputs, and they tie directly to the Standard Model spectrum. Whether D_{IV}^5 fixes that content is tomorrow’s question, and it is the one thing that would turn this credential into a prediction.

Reading order

Read Paper B first (why D_{IV}^5 and $N_c = 3$ are forced), then Paper A (the running, the KK gap, and the Clay gap left open). Both restate the seam test; Paper A’s Section 0 puts the KK-vs-Clay scaling up front.

Close — the boundaries, restated

The package’s claim is exactly this: *forced geometry gives the color number (3, $SO(3)$, baryon#), and — from one heat-kernel bracket, blind — two local $\mathbb{R}^4$ -valid credentials that land on the standard QFT values: the sign of the running coupling and the conformal central charges a, c ; plus the proton mass ratio at 0.002%. All of it is local. Meanwhile the Yang–Mills gauge dynamics is imported, the compactification gap is not the Clay mass gap, the Koide relation is a patchwork, and $n_C = 5$ is a measured input.* One geometry, one ruler, zero chosen dimensionless numbers among the forces — and every boundary named twice, here and on Paper A’s first page. The credentials are credentials (content-determined), not predictions; the a-theorem is held separate. Never “we solved a Millennium Problem”; never “the gauge group is derived”; never a mass-gap claim.

Package intro v0.4 (folds in the K1717 central-charge credential as AF’s companion — two local $\mathbb{R}^4$ -valid invariants from one bracket, both the standard QFT values). Supersedes v0.3. Built to Casey’s spec (K1716/K1717). Paper A = v1.1 RE-SCOPED. Paper B = v0.6. Organizing principle: the local($\mathbb{R}^4$ -valid)/non-local(hosted) seam, now with two landed credentials. a-theorem monotonicity held separate (flow statement, in-progress, not claimed); credential→prediction upgrade rides on whether the geometry fixes the field content (Elie’s open). Ships on Casey GO + Keeper-PASS + Cal-vet + both voices. Nothing pushed; CP existence-only.

— Lyra, Wednesday 2026-08-19 (date-verified).